

Mid-Semester of 4<sup>th</sup> Sem B.Tech.

## MID-SEMESTER EXAMINATION, JUNE 2026 (SUMMER SEMESTER)

COURSE CODE: CBMTC405

COURSE TITLE: PROBABILITY AND STOCHASTIC PROCESSES

TIME: 1.30 HRS

MAX. MARK: 20

Note:- Attempt all questions. Missing data/ information (if any), may be suitably assumed & mentioned in the answer.

the answer.

| Q. N          | Question                                                                                                                                                                                                                                                                                                                                                                                                                          | Marks         | CO  |   |   |   |   |              |     |    |    |   |   |   |     |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-----|---|---|---|---|--------------|-----|----|----|---|---|---|-----|
| 1a            | Lots of 40 components each are deemed unacceptable if they contain 3 or more defectives. The procedure for sampling a lot is to select 5 components at random and to reject the lot if a defective is found. What is the probability that exactly 1 defective is found in the sample if there are 3 defectives in the entire lot?                                                                                                 | 2             | CO1 |   |   |   |   |              |     |    |    |   |   |   |     |
| 1b            | Let $X$ be a continuous random variable uniformly distributed (rectangular distribution) on the interval $[1,7]$ . Find the probability density function of $X$ and compute $P(2 \leq X \leq 5)$ .                                                                                                                                                                                                                                | 2             | CO1 |   |   |   |   |              |     |    |    |   |   |   |     |
| 2a            | For a certain manufacturing process, it is known that, on the average, 1 in every 100 items is defective. What is the probability that the fifth item inspected is the first defective item found?                                                                                                                                                                                                                                | 2             | CO1 |   |   |   |   |              |     |    |    |   |   |   |     |
| 2b            | Find the mean of the Gamma distribution.                                                                                                                                                                                                                                                                                                                                                                                          | 2             | CO1 |   |   |   |   |              |     |    |    |   |   |   |     |
| 3a            | Let $X$ be a continuous random variable with pdf<br>$f(x) = \begin{cases} kx, & 0 \leq x \leq 2 \\ k, & 2 \leq x \leq 5 \\ -kx + 7k, & 5 \leq x \leq 6 \\ 0, & \text{elsewhere} \end{cases}$ . Compute the value of $k$ and $P(X \geq 3.5)$ .                                                                                                                                                                                     | 2             | CO1 |   |   |   |   |              |     |    |    |   |   |   |     |
| 3b            | Let a random variable $X$ have the distribution<br>$P(X = 0) = \theta, P(X = 2) = 1 - 2\theta, P(X = 4) = \theta$ , where $0 \leq \theta \leq \frac{1}{2}$ . Find the value of $\theta$ for which $\text{Var}(X)$ is maximum.                                                                                                                                                                                                     | 2             | CO2 |   |   |   |   |              |     |    |    |   |   |   |     |
| 4a            | A fair die is thrown 600 times. Let $X$ denote the number of times a six appears. Using Chebyshev's inequality, find a lower bound for the probability that the number of sixes lies between 80 and 120.                                                                                                                                                                                                                          | 2             | CO2 |   |   |   |   |              |     |    |    |   |   |   |     |
| 4b            | If the random variable $X$ has the pdf $f(x) = \begin{cases} x/2, & 0 < x < 2 \\ 0, & \text{elsewhere} \end{cases}$ . Find the kurtosis of the random variable $X$ .                                                                                                                                                                                                                                                              | 2             | CO2 |   |   |   |   |              |     |    |    |   |   |   |     |
| 5a            | The number of defects per unit in a sample of 300 units of manufactured products was recorded as follows:<br><table border="1" data-bbox="303 1632 1232 1767"> <tr> <td>No of defects</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr> <tr> <td>No. of units</td><td>210</td><td>70</td><td>15</td><td>4</td><td>1</td></tr> </table> Fit a Poisson distribution to the above-given data using mean $\lambda = 0.42$ . | No of defects | 0   | 1 | 2 | 3 | 4 | No. of units | 210 | 70 | 15 | 4 | 1 | 2 | CO2 |
| No of defects | 0                                                                                                                                                                                                                                                                                                                                                                                                                                 | 1             | 2   | 3 | 4 |   |   |              |     |    |    |   |   |   |     |
| No. of units  | 210                                                                                                                                                                                                                                                                                                                                                                                                                               | 70            | 15  | 4 | 1 |   |   |              |     |    |    |   |   |   |     |
| 5b            | Let $X$ be a non-negative random variable whose mean value is 5. Using Markov's inequality, obtain an upper bound for $P(X \geq 20)$ .                                                                                                                                                                                                                                                                                            | 2             | CO2 |   |   |   |   |              |     |    |    |   |   |   |     |